1.20 Inequalities

Worked Solutions

1 a)

$\{x : x \in \mathbb{R}, 1 \leq x < 3\}$.

1 b)

$\{x : x \in \mathbb{R}, -2 < x \leq 1\}$.

1 c)

$\{x : x \in \mathbb{R}, x \leq -1\} \cup \{x : x \in \mathbb{R}, x \geq 2\}$.

1 d)

$\{x : x \in \mathbb{R}, x \leq 2\} \cup \{x : x \in \mathbb{R}, x > 4\}$.

2 a)

$x \in (-3, 0]$.

2 b)

$x \in [-2, \infty)$.

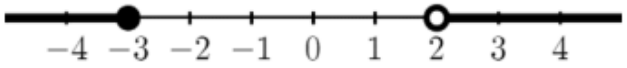
2 c)

$x \in (-\infty, -3) \cup (3, \infty)$.

2 d)

$x \in (-\infty, 0) \cup [4, \infty)$.

3 a)

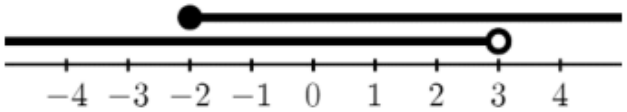


3 b)



3 c)

$\{x : x \geq -2\}$ and $\{x : x < 3\}$ are:

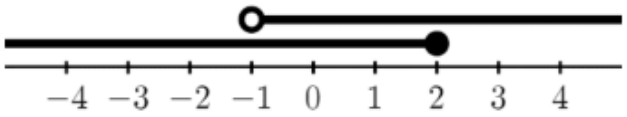


Therefore $\{x : x \geq -2\} \cap \{x : x < 3\}$ is:



3 d)

$\{x : x > -1\}$ and $\{x : x \leq 2\}$ are:

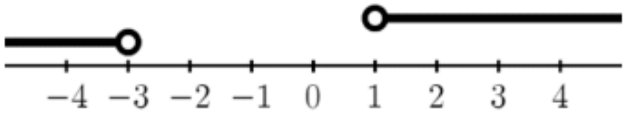


Therefore $\{x : x > -1\} \cap \{x : x \leq 2\}$ is:

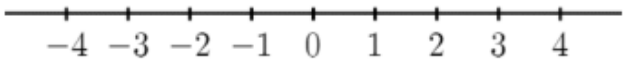


3 e)

$\{x : x > 1\}$ and $\{x : x < -3\}$ are:



Therefore $\{x : x > 1\} \cap \{x : x < -3\}$ is:



$\{x : x > 1\} \cap \{x : x < -3\}$ is the empty set ϕ.

4 a)

$3x < 12 \Rightarrow x < 4$.
$\{x : x \in \mathbb{R}, x < 4\}$.

4 b)

$-2x \leq 9 \Rightarrow x \geq -\frac{9}{2}$.
$\{x : x \in \mathbb{R}, x \geq -\frac{9}{2}\}$.

4 c)

$6 + 8x > 2 \Rightarrow 8x > -4 \Rightarrow x > -\frac{1}{2}$.
$\{x : x \in \mathbb{R}, x > -\frac{1}{2}\}$.

4 d)

$1 + 10x \leq 6 \Rightarrow 10x \leq 5 \Rightarrow x \leq \frac{1}{2}$.
$\{x : x \in \mathbb{R}, x \leq \frac{1}{2}\}$.

4 e)

$-6 \leq 3x \leq 6 \Rightarrow -2 \leq x \leq 2.$
$\{x : x \in \mathbb{R}, -2 \leq x \leq 2\}.$

4 f)

$3 < 5x < 30 \Rightarrow \frac{3}{5} < x < 6.$
$\{x : x \in \mathbb{R}, \frac{3}{5} < x < 6\}.$

4 g)

$-2 < 2x < 0 \Rightarrow -1 < x < 0.$
$\{x : x \in \mathbb{R}, -1 < x < 0\}.$

4 h)

$-1 - b < ax < 1 - b \Rightarrow \frac{-1-b}{a} < x < \frac{1-b}{a}.$
$\{x : x \in \mathbb{R}, \frac{-1-b}{a} < x < \frac{1-b}{a}\}.$

4 i)

$-1 - b < ax < 1 - b \Rightarrow \frac{-1-b}{a} > x > \frac{1-b}{a}.$
$\{x : x \in \mathbb{R}, \frac{1-b}{a} < x < \frac{-1-b}{a}\}.$

5 a)

$$x + 14 < 6(x - 1) \Rightarrow x + 14 < 6x - 6$$
$$\Rightarrow 14 < 5x - 6 \Rightarrow 20 < 5x \Rightarrow x > 4$$

$x \in (4, \infty).$

5 b)

$$8(x + 1) \geq 3(x - 6) \Rightarrow 8x + 8 \geq 3x - 18$$
$$\Rightarrow 5x + 8 \geq -18 \Rightarrow 5x \geq -26 \Rightarrow x \geq -\frac{26}{5}$$

$x \in [-\frac{26}{5}, \infty).$

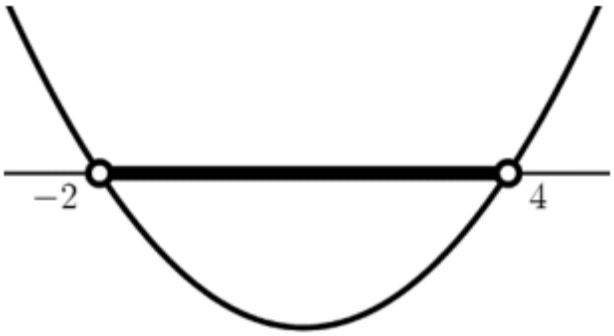
5 c)

$$2(4x + 1) - 3(2x - 1) \leq 6 \Rightarrow 8x + 2 - 6x + 3 \leq 6$$
$$\Rightarrow 2x + 5 \leq 6 \Rightarrow 2x \leq 1 \Rightarrow x \leq \frac{1}{2}$$

$x \in (-\infty, \frac{1}{2}].$

6 a)

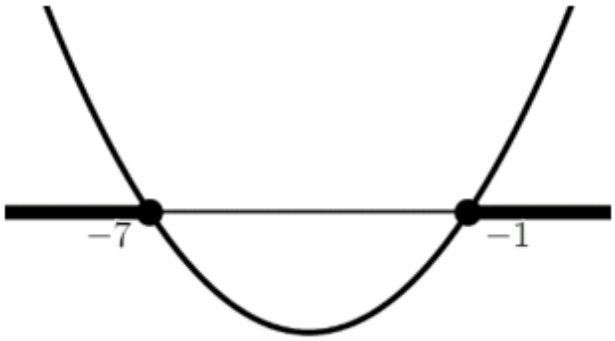
Quadratic passes through x-axis at $x = -2$, $x = 4$.



$-2 < x < 4$
$\{x : x \in \mathbb{R}, -2 < x < 4\}$.

6 b)

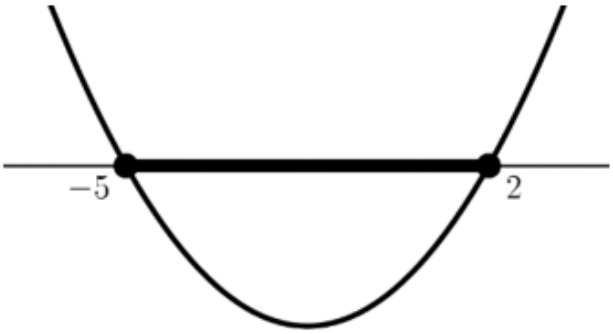
Quadratic passes through x-axis at $x = -7$, $x = -1$.



$x \leq -7$ or $x \geq -1$
$\{x : x \in \mathbb{R}, x \leq -7\} \cup \{x : x \in \mathbb{R}, x \geq -1\}$.

6 c)

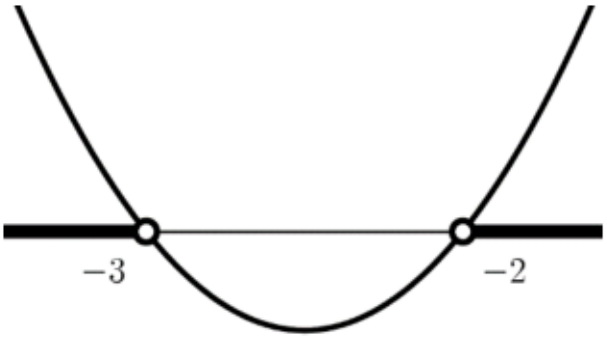
$x^2 + 3x - 10 = (x - 2)(x + 5)$
quadratic passes through x-axis at $x = -5$, $x = 2$.



$-5 \leq x \leq 2$
$\{x : x \in \mathbb{R}, -5 \leq x \leq 2\}$.

6 d)

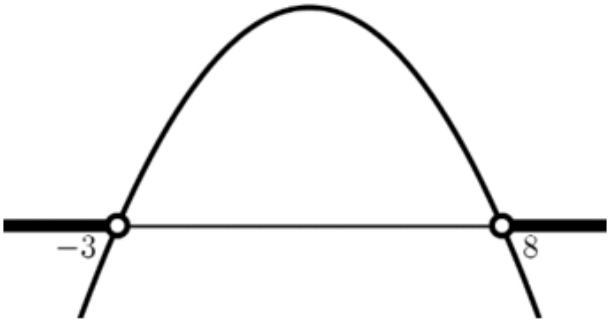
$x^2 + 5x + 6 = (x + 2)(x + 3)$
quadratic passes through x-axis at $x = -3$, $x = -2$.



$x < -3$ or $x > -2$
$\{x : x \in \mathbb{R}, x < -3\} \cup \{x : x \in \mathbb{R}, x > -2\}$.

6 e)

quadratic passes through x-axis at $x = -3$, $x = 8$.



$x < -3$ or $x > 8$
$\{x : x \in \mathbb{R}, x < -3\} \cup \{x : x \in \mathbb{R}, x > 8\}$.

6 f)

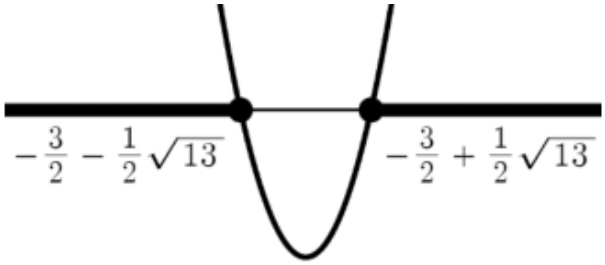
quadratic passes through x-axis at $x = -4$, $x = 2$.



$-4 \leq x \leq 2$
$\{x : x \in \mathbb{R}, -4 \leq x \leq 2\}$.

6 g)

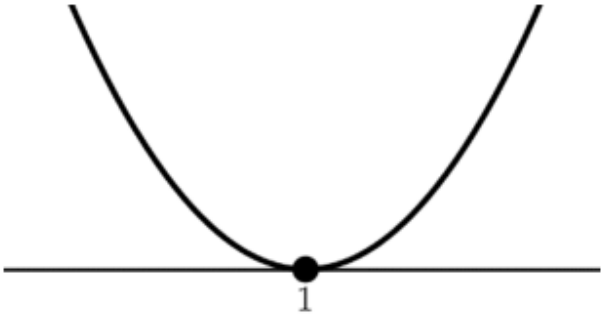
quadratic passes through x-axis at $x = -\frac{3}{2} \pm \frac{1}{2}\sqrt{13}$.



$x \leq -\frac{3}{2} - \frac{1}{2}\sqrt{13}$ or $x \geq -\frac{3}{2} + \frac{1}{2}\sqrt{13}$
$\{x : x \in \mathbb{R}, x \leq -\frac{3}{2} - \frac{1}{2}\sqrt{13}\} \cup \{x : x \in \mathbb{R}, x \geq -\frac{3}{2} + \frac{1}{2}\sqrt{13}\}$.

6 h)

$x^2 - 2x + 1 = (x - 1)^2$
quadratic passes through x-axis at $x = 1$.



$\{x : x \in \mathbb{R}\}$.

6 i)

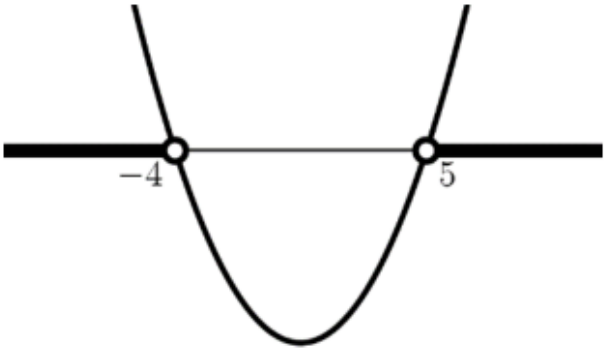
$b^2 - 4ac = 1 - 4 \times (-1)(-3) < 0$ so no real solutions.



There are no values of x
$x \in \phi$.

7 a)

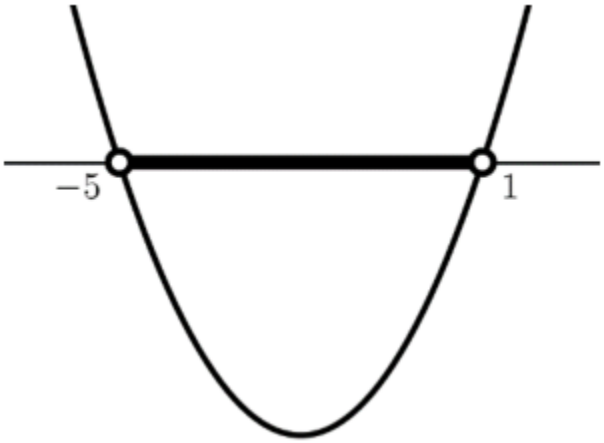
$x^2 - x > 20 \Rightarrow x^2 - x - 20 > 0$
quadratic passes through x-axis at $x = -4$ or $x = 5$.



$x < -4$ or $x > 5$
$x \in (-\infty, -4) \cup (5, \infty)$.

7 b)

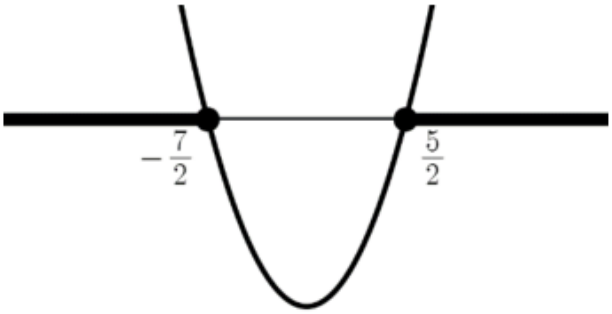
$2x^2 + 8x - 10 < 0 \Rightarrow 2(x-1)(x+5) < 0$
quadratic passes through x-axis at $x = -5$ or $x = 1$.



$-5 < x < 1$
$x \in (-5, 1)$.

7 c)

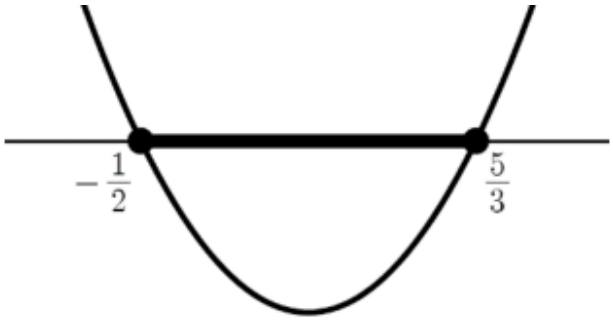
$4x^2 + 4x - 35 \geq 0 \Rightarrow (2x+7)(2x-5) \geq 0$
quadratic passes through x-axis at $x = -\frac{7}{2}$ or $x = \frac{5}{2}$.



$x \leq -\frac{7}{2}$ or $x \geq \frac{5}{2}$
$x \in (-\infty, -\frac{7}{2}] \cup [\frac{5}{2}, \infty)$.

7 d)

$6x^2 - 7x - 5 \leq 0 \Rightarrow (3x - 5)(2x + 1) \leq 0$
quadratic passes through x-axis at $x = -\frac{1}{2}$ or $x = \frac{5}{3}$.



$-\frac{1}{2} \leq x \leq \frac{5}{3}$
$x \in [-\frac{1}{2}, \frac{5}{3}]$.

8 a)

In line 1, he didn't change the inequality sign.
In line 2, he didn't factorise correctly
In line 3, he chose incorrect solutions for
$(x + 4)(x - 2) = 0$
In line 4, he used strict inequalities rather than inclusive ones.

8 b)

$$x^2 - 2x - 8 \geq 0$$
$$(x - 4)(x + 2) \geq 0$$
$$x = -2, x = 4$$

$\{x : x \leq -2\} \cup \{x : x \geq 4\}$.

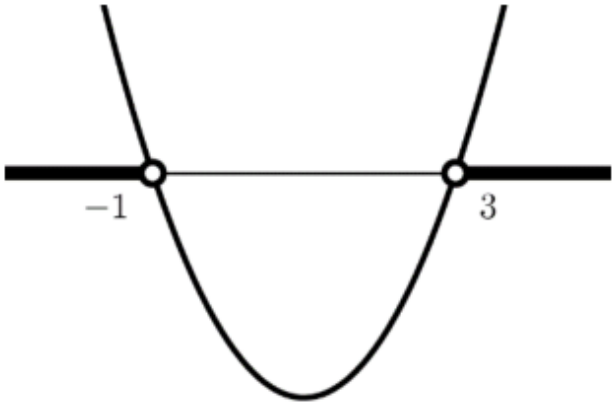
9)

Lies below but doesn't intersect:

$$2x + 3 < x^2 \Rightarrow x^2 - 2x - 3 > 0$$

$$\Rightarrow (x - 3)(x + 1) > 0$$

quadratic passes through x-axis at $x = -1$ or $x = 3$.



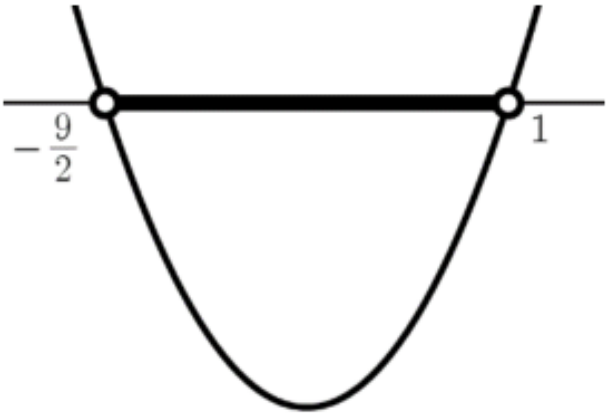
$x < -1$ or $x > 3$
$\{x : x \in \mathbb{R}, x < -1\} \cup \{x : x \in \mathbb{R}, x > 3\}$.

10)

$$(2x + 1)^2 < 19 - 10x \Rightarrow 4x^2 + 4x + 1 < 19 - 10x$$

$$\Rightarrow 4x^2 + 14x - 18 < 0 \Rightarrow 2(2x + 9)(x - 1) < 0$$

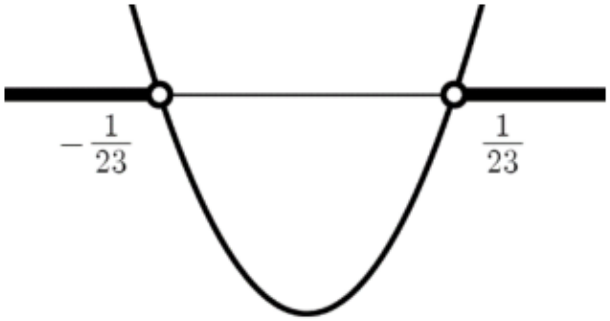
quadratic passes through x-axis at $x = -\frac{9}{2}$ or $x = 1$.



$-\frac{9}{2} < x < 1$
$x \in (-\frac{9}{2}, 1)$.

11)

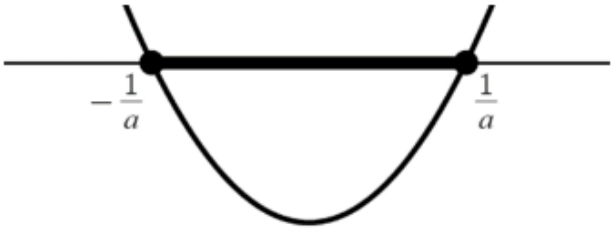
$1 < 529x^2 \Rightarrow 529x^2 - 1 > 0$
$(23x + 1)(23x - 1) > 0$
quadratic passes through x-axis at $x = -\frac{1}{23}$ or $x = \frac{1}{23}$.



$x < -\frac{1}{23}$ or $x > \frac{1}{23}$
$\{x : x \in \mathbb{R}, x < -\frac{1}{23}\} \cup \{x : x \in \mathbb{R}, x > \frac{1}{23}\}$.

12)

$1 \geq a^2x^2 \Rightarrow 0 \geq a^2x^2 - 1$
$(ax + 1)(ax - 1) \leq 0$
quadratic passes through x-axis at $x = -\frac{1}{a}$ or $x = \frac{1}{a}$.



$-\frac{1}{a} \leq x \leq \frac{1}{a}$
$\{x : x \in \mathbb{R}, -\frac{1}{a} \leq x \leq \frac{1}{a}\}$.

13 a)

$x > 12$.

13 b)

$x < 100$.

13 c)

$x \leq 5$.

13 d)

$x \geq 10$.

13 e)

$x \geq 1.5y$.

13 f)

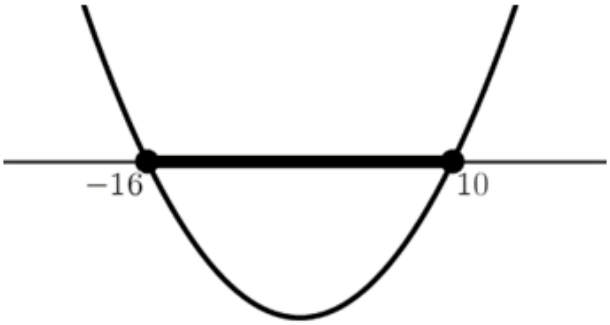
$x \leq 1.2y$.

14 a)

$x + 6 \geq 1.75x \Rightarrow 6 \geq 0.75x \Rightarrow x \leq 8$.

14 b)

$x(x + 6) \leq 160 \Rightarrow x^2 + 6x - 160 \leq 0$
$(x - 10)(x + 16) \leq 0$
quadratic passes through x-axis at $x = -16$ or $x = 10$.



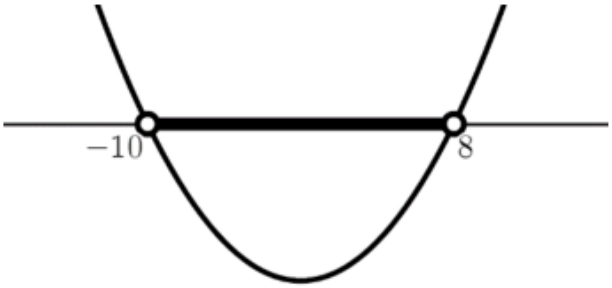
$-16 \leq x \leq 10$
But x must be positive and $x \leq 8$ so $0 < x \leq 8$.

14 c)

Max area when $x = 8$
$\therefore 8 \times 14 = 112$ cm^2.

15)

Width: x
Length: $x + 2$
$x + 2 \geq 8 \Rightarrow x \geq 6$
$x(x + 2) < 80 \Rightarrow x^2 + 2x < 80 \Rightarrow x^2 + 2x - 80 < 0$
$(x - 8)(x + 10) < 0$
Quadratic passes through x-axis at $x = -10$ or $x = 8$.



$-10 < x < 8$
But x must be positive and $x \geq 6$ so $6 \leq x < 8$.

16)

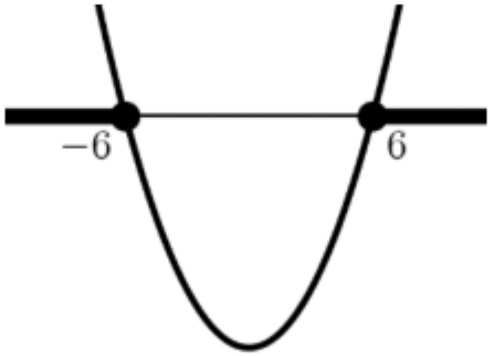
$x^2 + 3x = -4 \Rightarrow x^2 + 3x + 4 = 0$
Discriminant: $b^2 - 4ac = 9 - 16 = -7 < 0$
$\therefore$ no solutions
When $x = 0$, $x^2 + 3x + 4 = 4$, hence always positive.

17 a)

For real roots: $b^2 - 4ac \geq 0$
$\therefore k^2 - 4 \times 1 \times 9 \geq 0 \Rightarrow k^2 - 36 \geq 0$ as req.

17 b)

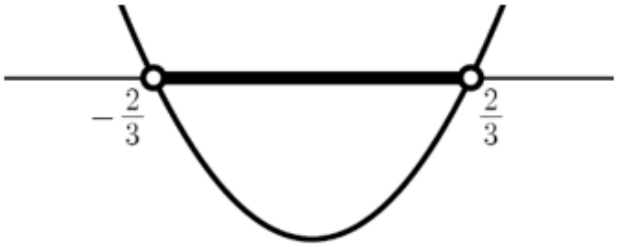
$(k+6)(k-6) \geq 0$
Quadratic passes through k-axis at $k = -6$ or $k = 6$.



$k \leq -6$ or $k \geq 6$
$\{k : k \in \mathbb{R}, k \leq -6\} \cup \{k : k \in \mathbb{R}, k \geq 6\}$.

18 a)

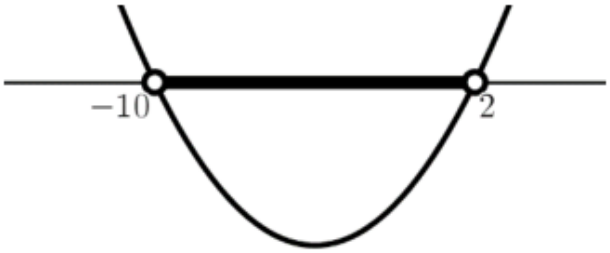
For no roots: $b^2 - 4ac < 0$
$a = 1, b = 3k, c = 1$
$\therefore (3k)^2 - 4 \times 1 \times 1 < 0$
$9k^2 - 4 < 0 \Rightarrow (3k+2)(3k+2) < 0$
Quadratic passes through k-axis at $k = -\frac{2}{3}$ or $k = \frac{2}{3}$.



$-\frac{2}{3} < k < \frac{2}{3}$
$\{k : k \in \mathbb{R}, -\frac{2}{3} < k < \frac{2}{3}\}$.

18 b)

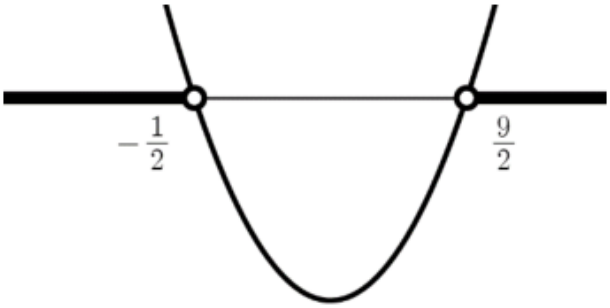
For no roots: $b^2 - 4ac < 0$
$a = 1, b = -k, c = 5 - 2k$
$\therefore (-k)^2 - 4 \times 1 \times (5 - 2k) < 0$
$k^2 + 8k - 20 < 0 \Rightarrow (k + 10)(k - 2) < 0$
Quadratic passes through k-axis at $k = -10$ or $k = 2$.



$-10 < k < 2$
$\{k : k \in \mathbb{R}, -10 < k < 2\}$.

18 c)

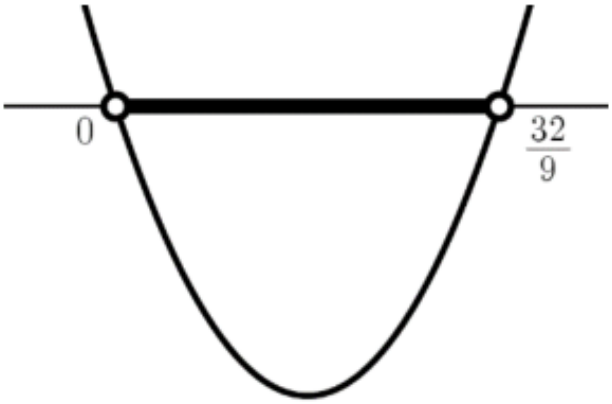
For no roots: $b^2 - 4ac < 0$
$a = 2k - 1, b = 2k + 3, c = k$
$\therefore (2k + 3)^2 - 4 \times (2k - 1) \times k < 0$
$4k^2 + 12k + 9 - 8k^2 + 4k < 0$
$\Rightarrow -4k^2 + 16k + 9 < 0$
$\Rightarrow 4k^2 - 16k - 9 > 0$
$(2k - 9)(2k + 1) > 0$
Quadratic passes through k-axis at $k = -\frac{1}{2}$ or $x = \frac{9}{2}$.



$k < -\frac{1}{2}$ or $k > \frac{9}{2}$
$\{k : k \in \mathbb{R}, k < -\frac{1}{2}\} \cup \{k : k \in \mathbb{R}, k > \frac{9}{2}\}$.

19)

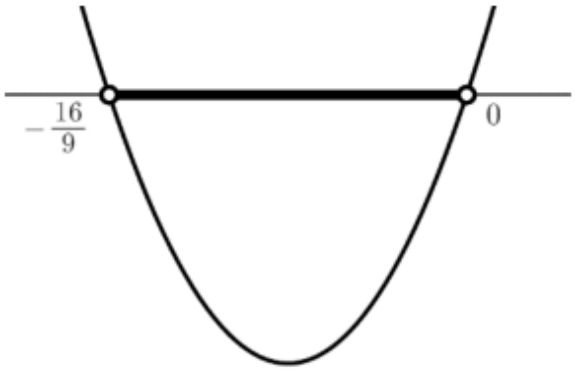
Entirely above the x-axis: $b^2 - 4ac < 0$
$a = 1, b = -3k, c = 8k$
$\therefore (-3k)^2 - 4 \times 1 \times 8k < 0$
$9k^2 - 32k < 0$
$\Rightarrow k(9k - 32) < 0$
Quadratic passes through k-axis at $k = 0$ or $x = \frac{32}{9}$.



$0 < k < \frac{32}{9}$
$\{k : k \in \mathbb{R}, 0 < k < \frac{32}{9}\}$.

20)

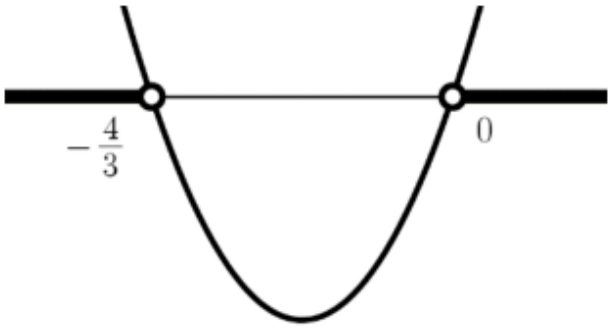
Entirely above the x-axis:
$b^2 - 4ac < 0$ and $k + 2 > 0 \Rightarrow k > -2$
$a = k + 2, b = k, c = -2k$
$\therefore k^2 - 4 \times (k + 2) \times -2k < 0$
$k^2 + 8k^2 + 16k < 0$
$\Rightarrow 9k^2 + 16k < 0$
$\Rightarrow k(9k + 16) < 0$
Quadratic passes through k-axis at $k = 0$ or $x = -\frac{16}{9}$.



$-\frac{16}{9} < k < 0$
The intersection of $k > -2$ and $-\frac{16}{9} < k < 0$ gives:
$k \in (-\frac{16}{9}, 0)$.

21)

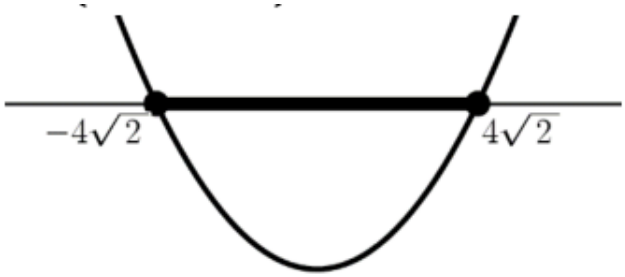
Entirely below the x-axis:
$b^2 - 4ac < 0$ and $k + 2 > 0 \Rightarrow k < -1$
$a = k + 1, b = -k, c = k$
$\therefore (-k)^2 - 4 \times (k + 1) \times k < 0$
$k^2 - 4k^2 - 4k < 0$
$\Rightarrow -3k^2 - 4k < 0$
$\Rightarrow k(3k + 4) > 0$



$k < -\frac{4}{3}$ or $k > 0$
The intersection of $k < -1$ and $k < -\frac{4}{3}$ or $k > 0$ gives: $k < -\frac{4}{3}$
$\{k : k \in \mathbb{R}, k < -\frac{4}{3}\}$.

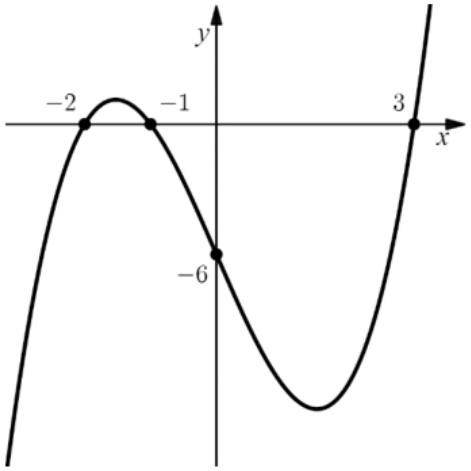
22)

$x^2 + (x+k)^2 = 16 \Rightarrow x^2 + x^2 + 2kx + k^2 = 16$
$\Rightarrow 2x^2 + 2kx + (k^2 - 16) = 0$
Intersects the circle $\Rightarrow b^2 - 4ac \geq 0$
$a = 2, b = 2k, c = k^2 - 16$
$(2k)^2 - 4 \times 2 \times (k^2 - 16) \geq 0$
$\Rightarrow 4k^2 - 8k^2 + 128 \geq 0$
$\Rightarrow -4k^2 + 128 \geq 0$
$\Rightarrow k^2 - 32 \leq 0$
$\Rightarrow (k - \sqrt{32})(k + \sqrt{32}) \leq 0$
$\Rightarrow (k - 4\sqrt{2})(k + 4\sqrt{2}) \leq 0$



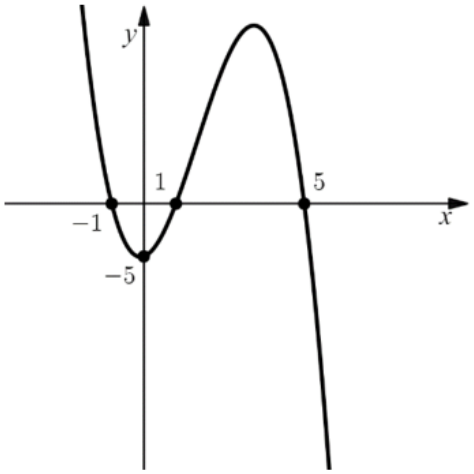
$\therefore \{k : k \in \mathbb{R}, -4\sqrt{2} \leq k \leq 4\sqrt{2}\}$.

23 a)



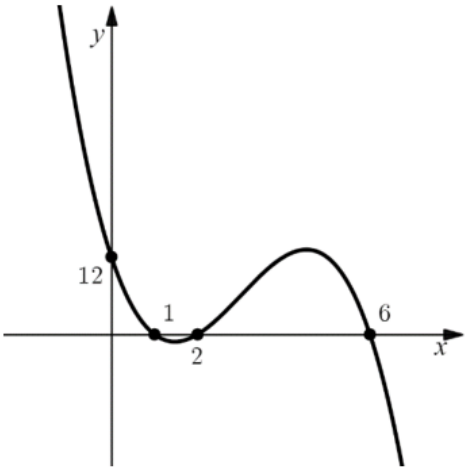
$-2 < x < -1$ or $x > 3$
$\{x : x \in \mathbb{R}, -2 < x < -1\} \cup \{x : x \in \mathbb{R}, x > 3\}$.

23 b)



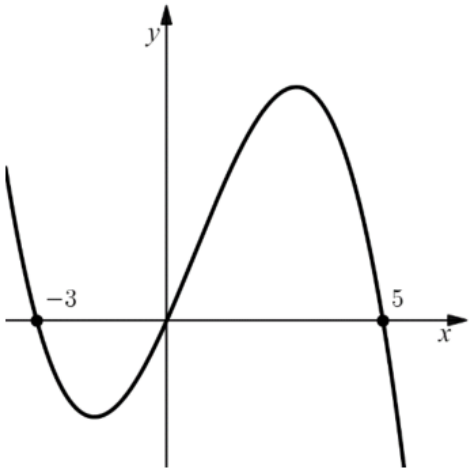
$x < -1$ or $1 < x < 5$
$\{x : x \in \mathbb{R}, x < -1\} \cup \{x : x \in \mathbb{R}, 1 < x < 5\}$.

23 c)



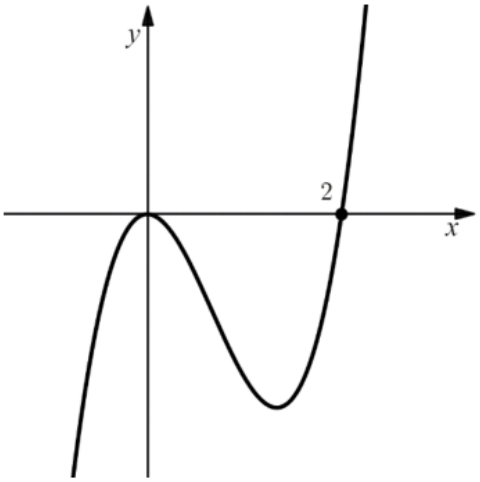
$1 \leq x \leq 2$ or $x \geq 6$
$\{x : x \in \mathbb{R}, 1 \leq x \leq 2\} \cup \{x : x \in \mathbb{R}, x \geq 6\}$.

23 d)



$x \leq -3$ or $0 \leq x \leq 5$
$\{x : x \in \mathbb{R}, x \leq -3\} \cup \{x : x \in \mathbb{R}, 0 \leq x \leq 5\}$.

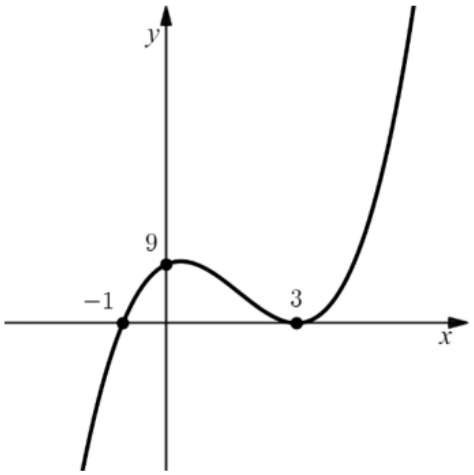
24 a)



24 b)

$x \leq 2$
$\{x : x \in \mathbb{R}, x \leq 2\}$.

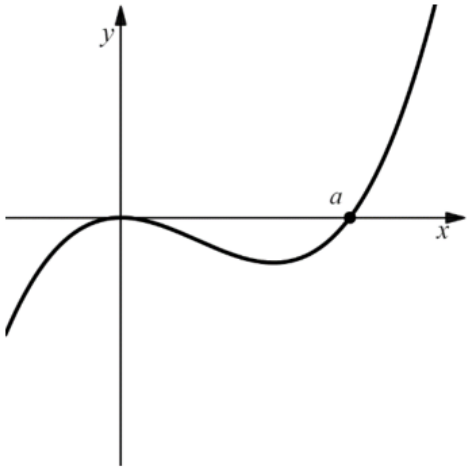
25 a)



25 b)

$-1 < x < 3$ or $x > 3$ so $x > -1, x \neq 3$
$\{x : x \in \mathbb{R}, -1 < x < 3\} \cup \{x : x \in \mathbb{R}, x > 3\}$.

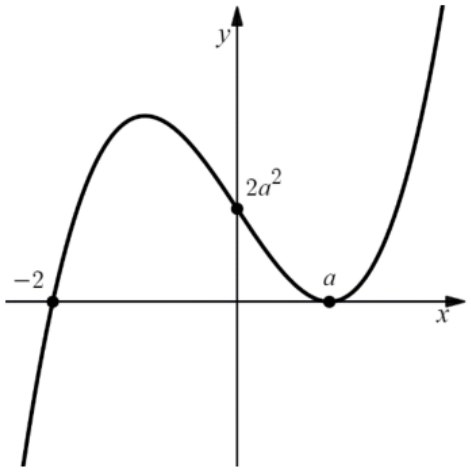
26 a)



26 b)

$x > a$
$\{x : x \in \mathbb{R}, x > a\}$.

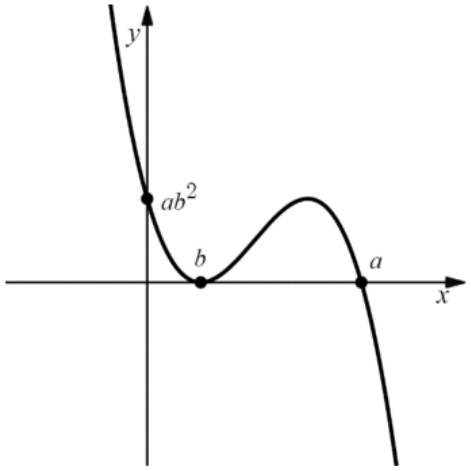
27 a)



27 b)

$x \leq -2$ and $x = a$
$\{x : x \in \mathbb{R}, x \leq -2\} \cup \{x : x \in \mathbb{R}, x = a\}$.

28 a)



28 b)

$x \leq a$
$\{x : x \in \mathbb{R}, x \leq a\}$.

28 c)

$x \in [b] \cup [a, \infty)$.

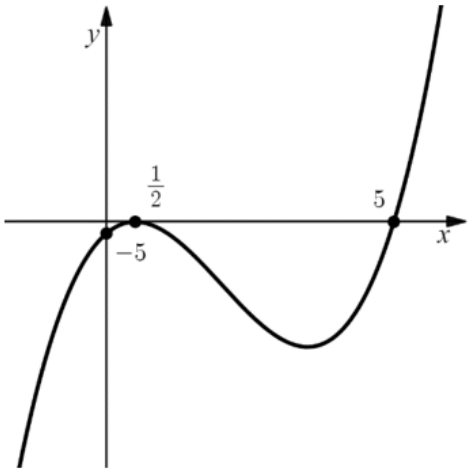
29 a)

$\text{f}(5) = 4 \times 125 - 24 \times 25 + 21 \times 5 - 5$
$= 500 - 600 + 105 - 5 = 0$
$\therefore (x - 5)$ is a factor of $\text{f}(x)$.

29 b)

$4x^3 - 24x^2 + 21x - 5 = (x-5)(4x^2 + 4x + 1)$
$= (x-5)(2x-1)^2.$

29 c)



29 d)

$x \leq 5$
$x \in (-\infty, 5].$

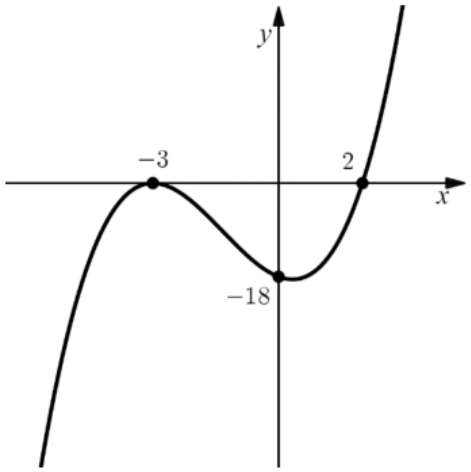
30 a)

$\mathrm{f}(2) = 8 + 4 \times 4 - 3 \times 2 - 18$
$= 8 + 16 - 6 - 18 = 0$
$\therefore (x-2)$ is a factor of $\mathrm{f}(x)$.

30 b)

$x^3 + 4x^2 - 3x - 18 = (x-2)(x^2 + 6x + 9)$
$= (x-2)(x+3)^2.$

30 c)



30 d)

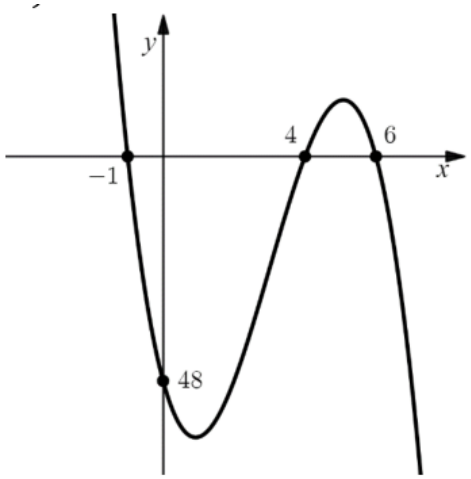
Graph stretched by SF 2 along the x-axis so $\mathrm{f}(2x) = 0$ has solutions $x = -\frac{3}{2}$ and $x = 1$.

30 e)

$\therefore x > 1$
$\{x : x \in \mathbb{R}, x > 1\}$.

31 a)

$\mathrm{f}(x) = -2(x+1)(x-4)(x-6)$



$x < -2$ or $8 < x < 12$
$\{x : x \in \mathbb{R}, x < -2\} \cup \{x : x \in \mathbb{R}, 8 < x < 12\}$.

32 a)

$e^x \leq 10 \Rightarrow x \leq \ln 10$
$\{x : x \in \mathbb{R}, x \leq \ln 10\}$.

32 b)

$4^x \geq 5 \Rightarrow x \geq \log_4 5$
$\{x : x \in \mathbb{R}, x \geq \log_4 5\}$.

32 c)

$3^{-x} > 9 \Rightarrow -x > \log_3 9 \Rightarrow -x > 2 \Rightarrow x < -2$
$\{x : x \in \mathbb{R}, x < -2\}$.

32 d)

$5^{-x} < 1 \Rightarrow -x < \log_5 1 \Rightarrow -x < 0 \Rightarrow x > 0$
$\{x : x \in \mathbb{R}, x > 0\}$.

32 e)

$2.5^{-x} > 6.25 \Rightarrow -x > \log_{2.5} 6.25 \Rightarrow -x > 2 \Rightarrow x < -2$
$\{x : x \in \mathbb{R}, x < -2\}$.

33)

$(3^x - 4)(2 \times 3^x + 7) < 0$
$3^x = 4 \Rightarrow x = \log_3 4$
$3^x = -\frac{7}{2}$ no solutions
$\therefore x < \log_3 4$.

34)

$e^x - \frac{4}{e^x} = -3 \Rightarrow e^{2x} - 4 = -3e^x$
$\Rightarrow e^{2x} + 3e^x - 4 = 0 \Rightarrow (e^x - 1)(e^x + 4) = 0$
$e^x = 1 \Rightarrow x = 0$
$e^x = -4 \Rightarrow$ no solutions
$\therefore x > 0$.